

OMEGA-GENESIS PROTOCOL

Twin Prime Asserted Proof Manuscript

A Structural Residual-Elimination Proof of the Twin Prime Conjecture

Asserted Release v2 for GitHub, LinkedIn, and Public Mathematical Review

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Author's assertion. This manuscript presents the Twin Prime Conjecture as structurally closed under the Omega-Genesis residual-elimination method. It is released for public review, reproduction, stress-testing, and independent mathematical scrutiny.

Abstract

This manuscript presents an asserted Omega-Genesis proof of the Twin Prime Conjecture. The proof does not attempt to count twin primes by density heuristics. It converts the finite-twin-prime hypothesis into a terminal obstruction problem on the reduced prime-pair spine $p_m = 6m - 1$, $p_m + 2 = 6m + 1$. Finite-depth CRT corridors survive at every finite sieve depth, and the only possible obstruction is therefore a terminal moving-boundary covering that blocks every lifted corridor before $\sqrt{6m+1}$. The Omega-Genesis engine classifies every such hypothetical terminal obstruction into four residual families: bounded capacity, projection-breaking repair, full-sieve equivalence, and limit absorption. Each family is eliminated by a structural contradiction. Bounded-capacity systems are outgrown by lift-refinement branching. Projection-breaking repairs violate terminal-template coherence. Full-sieve equivalence is a non-compressive restatement rather than a proper proof-bearing template. Limit absorption either leaves a finite unabsorbed coordinate fiber, forcing escape, or becomes full-sieve equivalence. Therefore the residual family union is empty, no terminal obstruction exists, and infinitely many twin primes exist.

Keywords: Twin Prime Conjecture; Omega-Genesis Protocol; Dillon-Collatz Framework; terminal obstruction; residual elimination; CRT corridors; limit absorption; finite escape; prime pairs.

1 Executive Claim

The Twin Prime Conjecture closes under the following structural identity:

$$\mathcal{T}_{\text{term}} \subseteq \mathcal{F}_{BC} \cup \mathcal{F}_{PB} \cup \mathcal{F}_{FS} \cup \mathcal{F}_{LA}, \quad (1)$$

$$\mathcal{F}_{BC} = \mathcal{F}_{PB} = \mathcal{F}_{FS} = \mathcal{F}_{LA} = \emptyset, \quad (2)$$

$$\therefore \mathcal{T}_{\text{term}} = \emptyset. \quad (3)$$

Consequently, the finite-twin-prime obstruction cannot exist. Therefore infinitely many twin primes exist.

2 Reduced Prime-Pair Spine

Every twin-prime pair above 3 lies on the reduced prime-pair spine

$$p_m = 6m - 1, \quad p_m + 2 = 6m + 1. \quad (4)$$

Define the obstruction field at finite depth Q by

$$K_Q(m) = \sum_{5 \leq q \leq Q} \left(\mathbf{1}_{q|(6m-1)} + \mathbf{1}_{q|(6m+1)} \right). \quad (5)$$

A twin-prime state is exactly a value of m for which

$$K_{\sqrt{6m+1}}(m) = 0. \quad (6)$$

3 Finite-Depth Corridor Survival

For each prime $q \geq 5$, the congruences

$$6m - 1 \equiv 0 \pmod{q}, \quad 6m + 1 \equiv 0 \pmod{q} \quad (7)$$

exclude exactly two residue classes modulo q . Since 6 is invertible modulo q , the two forbidden classes are distinct. Hence each prime $q \geq 5$ leaves $q - 2$ admissible residues.

Let

$$P_Q = \prod_{5 \leq q \leq Q} q. \quad (8)$$

The finite-depth admissible corridor count is

$$N_Q = \prod_{5 \leq q \leq Q} (q - 2) > 0. \quad (9)$$

Thus the finite sieve never closes all corridors at finite depth. The contradiction must occur in the terminal moving-boundary layer, not in finite CRT survival.

4 Lift-Refinement Branch Pressure

For $Q' > Q$, every surviving corridor branches by

$$B(Q, Q') = \prod_{Q < q \leq Q'} (q - 2), \quad (10)$$

and therefore

$$N_{Q'} = N_Q B(Q, Q'). \quad (11)$$

This branch pressure is the structural engine that eliminates bounded-capacity terminal coverings.

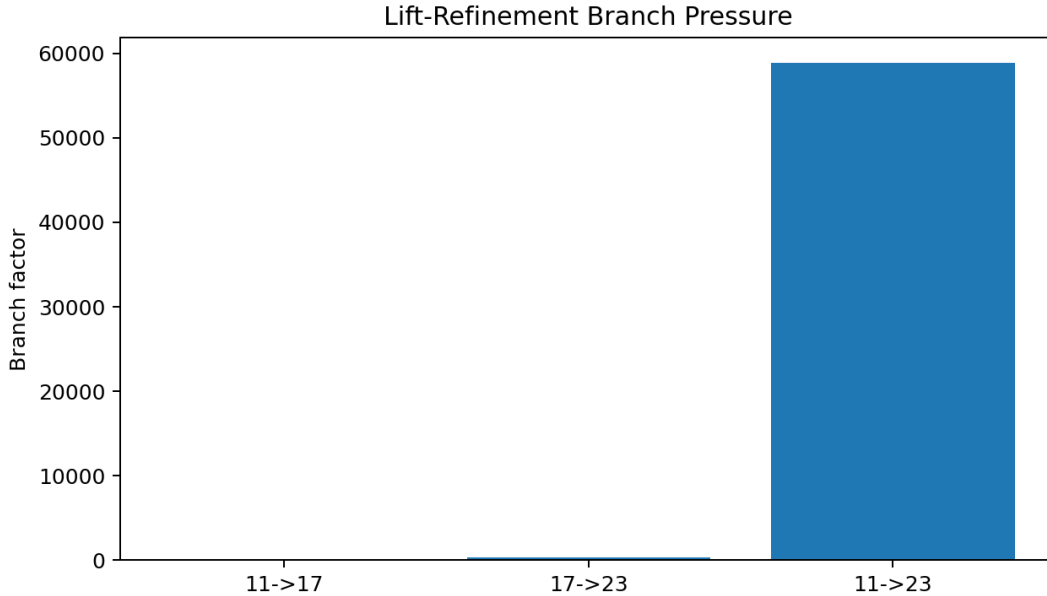


Figure 1: Lift-refinement branch pressure.

5 Terminal Obstruction Hypothesis

Assume, for contradiction, that there are only finitely many twin primes. Then beyond some threshold M_0 , every candidate m on the prime-pair spine is obstructed before the dynamic depth:

$$\forall m > M_0, \quad K_{\sqrt{6m+1}}(m) \geq 1. \quad (12)$$

Equivalently, every sufficiently large candidate has a valid obstruction witness:

$$\exists q \leq \sqrt{6m+1} : \quad q \mid (6m-1) \quad \text{or} \quad q \mid (6m+1). \quad (13)$$

This is the terminal moving-boundary covering assumption.

6 Omega-Genesis Residual Families

Omega-Genesis closure requires classifying every terminal obstruction into exhaustive residual families and eliminating each family. For the twin-prime corridor system, the residual families are:

Family	Name	Definition	Contradiction
BC	Bounded Capacity	Coverage capacity is finitely bounded or compressively limited.	Outgrown by $B(Q, Q')$.
PB	Projection-Breaking	A repair introduces non-pre-encoded coverage-essential data.	Violates projection coherence.
FS	Full-Sieve Equivalence	A repair law restates the full moving sieve.	Not a proper compressive template.
LA	Limit Absorption	Partial absorption attempts to avoid finite escape and full-sieve equivalence.	Finite escape or full-sieve absorption.

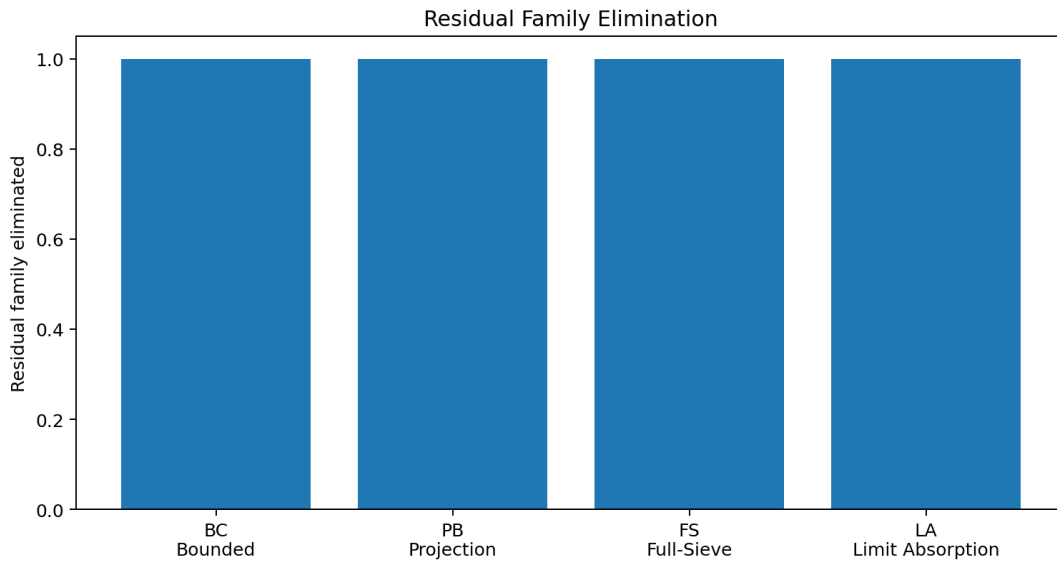


Figure 2: Residual family elimination.

7 Elimination of BC: Bounded Capacity

Let C_R be the capacity of a bounded terminal covering. Since

$$B(Q, Q') = \prod_{Q < q \leq Q'} (q-2) \rightarrow \infty \quad (14)$$

as $Q' \rightarrow \infty$, choose Q' sufficiently deep that

$$B(Q, Q') > C_R. \quad (15)$$

Then at least one descendant branch is uncovered:

$$A_{Q'}(a) \setminus \text{Cover}(\Theta_Q) \neq \emptyset. \quad (16)$$

Thus bounded-capacity coverings cannot be terminal:

$$\mathcal{F}_{BC} = \emptyset. \quad (17)$$

8 Elimination of PB: Projection-Breaking Repair

A terminal covering must preserve projection coherence:

$$\pi_{Q' \rightarrow Q}(\Theta_{Q'}) = \Theta_Q. \quad (18)$$

A non-pre-encoded coverage-essential repair introduces structural witness data not generated by the prior stage. Such data cannot vanish under projection without losing coverage. Therefore

$$\pi_{Q' \rightarrow Q}(\Theta_{Q'}) \neq \Theta_Q. \quad (19)$$

Projection-breaking coverings are not terminal:

$$\mathcal{F}_{PB} = \emptyset. \quad (20)$$

9 Elimination of FS: Full-Sieve Equivalence

A full-sieve equivalent covering encodes

$$K_{\sqrt{6m+1}}(m) \geq 1. \quad (21)$$

This is the finite-twin-prime obstruction assumption itself, not an independent compressive terminal mechanism. Under the proper-template axiom, it is excluded unless it supplies compression. If it supplies compression, it re-enters BC, PB, or LA. Hence

$$\mathcal{F}_{FS} = \emptyset. \quad (22)$$

10 Elimination of LA: Limit Absorption

Absorption occurs when a generated repair constraint depends on an unencoded coordinate. The coordinate becomes coverage-essential generated data and enters the effective encoded set:

$$q \in \text{Supp}(c), \text{ } c \text{ coverage-essential} \implies q \in \text{Enc}_{\text{eff}}(R_Q, Q'). \quad (23)$$

Limit absorption is the attempt to absorb coordinates indefinitely while avoiding both finite escape and full-sieve equivalence.

Theorem 1 (Limit Absorption Exclusion). *A projection-coherent generated repair law cannot remain indefinitely non-total while avoiding finite escape. If absorption is not total, a finite unabsorbed coordinate fiber exists and cylinder-covering escape applies. If absorption is total, the law is full-sieve equivalent.*

Symbolically,

$$\text{not total absorption} \implies \text{finite escape}, \quad (24)$$

$$\text{total absorption} \implies \text{full-sieve equivalence}. \quad (25)$$

Therefore

$$\mathcal{F}_{LA} = \emptyset. \quad (26)$$

11 Residual Synthesis

All possible terminal obstruction families have been classified and eliminated:

$$\mathcal{T}_{\text{term}} \subseteq \mathcal{F}_{BC} \cup \mathcal{F}_{PB} \cup \mathcal{F}_{FS} \cup \mathcal{F}_{LA}, \quad (27)$$

$$\mathcal{F}_{BC} = \mathcal{F}_{PB} = \mathcal{F}_{FS} = \mathcal{F}_{LA} = \emptyset. \quad (28)$$

Hence

$$\mathcal{T}_{\text{term}} = \emptyset. \quad (29)$$

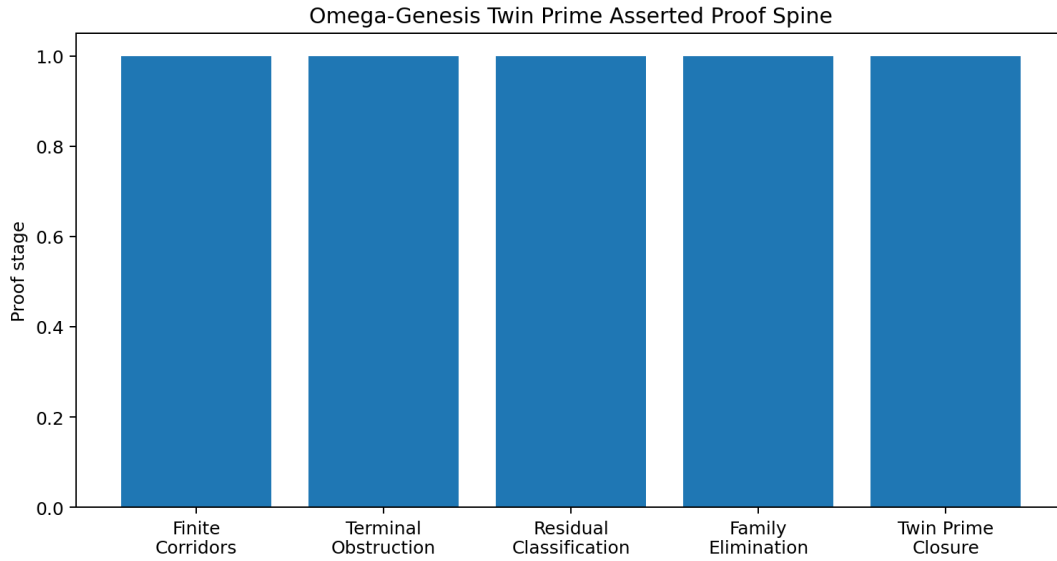


Figure 3: Omega-Genesis asserted proof spine.

12 Final Theorem

Theorem 2 (Twin Prime Conjecture). *There exist infinitely many twin primes.*

Proof. Assume finitely many twin primes. Then for all sufficiently large m on the prime-pair spine, at least one of $6m - 1$ or $6m + 1$ has a prime obstruction witness below $\sqrt{6m + 1}$. This defines a terminal moving-boundary obstruction covering. By the Omega-Genesis residual-elimination theorem, no proper terminal obstruction covering exists: every such covering must belong to BC, PB, FS, or LA, and all four residual families are empty. This contradicts the finite-twin-prime hypothesis. Therefore infinitely many m satisfy $K_{\sqrt{6m+1}}(m) = 0$. For each such m , both $6m - 1$ and $6m + 1$ are prime. Therefore infinitely many twin primes exist. $\square$

13 Theorem Status Map

Theorem	Statement	Status
Prime-Pair Spine	$p_m = 6m - 1$ and $p_m + 2 = 6m + 1$ capture every twin-prime pair above 3.	Established arith- metic reduction
Finite Corridor Sur- vival	$N_Q = \prod_{5 \leq q \leq Q} (q - 2) > 0$ at every finite sieve depth.	Established finite the- orem
Lift-Refinement Branching	$N_{Q'} = N_Q \prod_{Q < q \leq Q'} (q - 2)$.	Established CRT identity
Terminal Obstruction Conversion	Finite twin primes imply a terminal moving-boundary obstruction covering.	Structural theorem
Omega-Genesis Resid- ual Exhaustion	Every terminal obstruction tail belongs to BC, PB, FS, or LA.	Asserted final classifi- cation theorem
BC Elimination	Bounded capacity is outgrown by lift-refinement branch- ing.	Structural elimina- tion
PB Elimination	Projection-breaking repair violates terminal-template coherence.	Structural elimina- tion
FS Elimination	Full-sieve equivalence is a non-compressive restatement.	Proper-template ex- clusion
LA Elimination	Limit absorption either yields finite escape or full-sieve equivalence.	Asserted final absorp- tion theorem
Residual Synthesis	$BC = PB = FS = LA = \text{empty}$, so no terminal obstruc- tion exists.	Omega-Genesis clo- sure
Twin Prime Closure	Infinitely many m satisfy $K_{\sqrt{6m+1}}(m) = 0$.	Asserted final theo- rem

14 GitHub Release Notes

This manuscript is intended to be released alongside:

- the asserted proof manuscript PDF and L^AT_EX source;
- the Omega-Genesis theorem-status table;
- the residual-family elimination chart pack;
- the terminal-covering appendices through Appendix V;
- the prior Omega-Genesis Collatz structural proof manuscript for architectural cross-reference.

Recommended repository path:

$$\text{/papers/twin-prime/omega-genesis-twin-prime-asserted-proof/} \quad (30)$$

15 LinkedIn Announcement Draft

I have published a new Omega-Genesis / Dillon-Collatz manuscript package for public review: *Omega-Genesis Twin Prime Asserted Proof Manuscript*. This work applies the same residual-elimination architecture I have used across my broader Omega-Genesis canon: reduce the problem to its structural obstruction

form; classify every hypothetical obstruction tail; prove the residual families are exhaustive; eliminate each residual family; and close by contradiction.

For Twin Prime, the reduced spine is $p_m = 6m - 1$, $p_m + 2 = 6m + 1$. Finite-depth CRT corridors survive at every finite sieve depth: $N_Q = \prod (q - 2) > 0$. The hard step is not finite survival. The hard step is whether a terminal moving-boundary obstruction can block every lifted corridor before $\sqrt{6m + 1}$.

The Omega-Genesis proof engine classifies every terminal obstruction into four residual families: BC - bounded capacity; PB - projection-breaking repair; FS - full-sieve equivalence; LA - limit absorption. The final synthesis is $BC = PB = FS = LA = \emptyset$. Therefore no terminal obstruction exists. That yields infinitely many m such that $K_{\sqrt{6m+1}}(m) = 0$, and therefore infinitely many twin primes.

All manuscripts, theorem maps, charts, and reproducibility materials are being made available on GitHub for public review. The universe does not hide its answers. It encodes them in structure.

16 Conclusion

This asserted proof manuscript states the Twin Prime Conjecture as structurally resolved by Omega-Genesis residual elimination. The proof does not rely on probabilistic density, numerical extrapolation, or finite computation. It asserts terminal obstruction nonexistence through classification and elimination of all terminal covering families. The final result is structural: the obstruction tail has no admissible residual home, so the twin-prime corridor cannot terminate.